

Neural Space Time Memory

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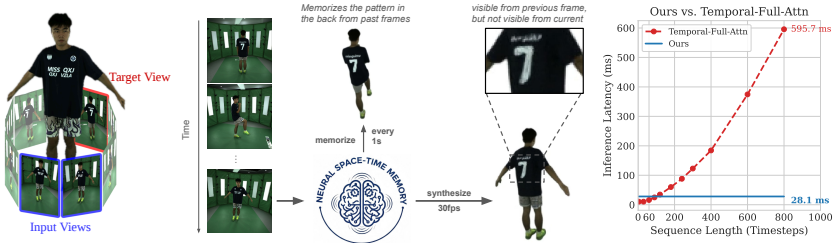


Fig. 1: We propose Neural Space Time Memory, a novel method that takes multi-view video streams as input, and performs novel view synthesis in *real-time*. The key feature of our method is being able to memorize *minute-long* past content. As shown in the figure, even if the back of the person is only visible from the first frame and the person has moved, we are able to faithfully and efficiently render the back view by incorporating both the current frame and the memory content.

Abstract. Novel view synthesis from dynamic streaming video faces a fundamental trade-off: maintaining a persistent, long-horizon memory to reconstruct temporarily occluded regions while operating under strict real-time constraints. While Test-Time Training (TTT) offers a powerful memory mechanism, standard models mandate gradient-based memory updates at every frame to adapt to the changing motion in dynamic scenes. This computational overhead from heavy memory updates precludes real-time application and can lead to instability over extended contexts. Recognizing that memory updates are considerably more demanding than memory application and that video content is largely redundant, we propose to decouple the frequencies of these two process. Our approach performs periodic memory updates while applying the memory on a per-frame basis, utilizing cross-view attention to manage deformations between the prior memory state and the current frame. To lock in the historical context, we introduce two critical mechanisms: an auxiliary *Memory Loss* that forces persistent internalization of the scene, and a *Memory Caching* strategy that regularizes active weights against catastrophic drift. Extensive evaluations on unconstrained, dynamic human motion demonstrate that our method exhibits state-of-the-art performance, and successfully demonstrates minute-scale online memorization while operating in real-time.

1 Introduction

Humans possess an innate space-time memory. When observing dynamic environments, the continuous visual stream we process is inherently incomplete at any given moment due to frequent occlusions and limited fields of view. Yet, our brains effortlessly reason through motion and object permanence, seamlessly fusing partial observations over time to imagine the current state of temporarily hidden entities. Computationally realizing such a real-time space-time memory system would have many downstream applications, but it has proven elusive due to the inability to maintain a coherent, persistent memory of unconstrained dynamic scenes over long time horizons, without sacrificing real-time performance. Dynamic *Novel View Synthesis (NVS)* serves as an ideal, rigorous stress test for such neural memory due to its reliance on partial inputs and the requirement to fuse and synthesize from this historical context. Therefore, in this paper, we tackle the challenge of real-time dynamic NVS from streaming videos. Our objective is to build a system capable of storing persistent, minute-scale historical context to perform grounded synthesis of momentarily occluded regions, all while operating under strict real-time constraints.

In this paper, we present a memory system based on *Test-Time Training (TTT)*. We selected TTT for its expressive memory capacity and, crucially, its linear scalability – bypassing the prohibitive quadratic computational cost of full self-attention used in typical feed-forward NVS methods (Fig. 1). However, while TTT has proven highly effective for static 3D environments, deploying it on continuous, unconstrained dynamic scenes poses several challenges, as follows.

Speed. The efficiency of recent TTT models is from large-chunk processing [64]. Naïvely adapting this to dynamic environments requires updating its memory at every frame to adapt to the changing motion. The significant computational overhead from heavy memory updates, however, precludes real-time application. Recognizing that memory updates are considerably more demanding than memory applications (Fig. 5)—and that dynamic video content is largely redundant—we propose a novel framework that decouples the frequencies of these two processes. Our approach performs infrequent periodic, heavy memory updates to capture novel states (*Memorization Step*), while applying the memory continuously on a per-frame basis (*Synthesis Step*). During the Synthesis Step, the attention module attends across current input and target views to capture ongoing motion, while querying the memory module for past context. Together, they resolve dynamic deformations between historical states and present observations, unlocking amortized real-time performance.

Memory Persistence and Supervision Balance. Because disocclusions in streaming video provide only a sparse supervisory signal, jointly training the memory module and the cross-view attention module allows the model to take a shortcut. The network learns to over-rely on the attention’s processing of the current, incomplete observation, effectively bypassing the memory module and collapsing to a stateless prior (Sec. 4.4). To resolve this, we disentangle the training objectives and propose an auxiliary *memory loss*. During the *Memorization Step*, we query the memory using only the target view camera rays,

deliberately preventing any attention over the current input views. By isolating the readout, we force the memory module to reconstruct the scene entirely from its internalized historical context. Then, during the *Synthesis Step*, we combine the current motion from the cross-view attention and the past context from the memory module to synthesize the final novel view. By explicitly supervising the memory’s independent reconstruction capabilities, we force the fast-weights to persistently internalize the scene context.

Long-Term Stability. Continuous memory updates in TTT can lead to instability over long horizons. Our periodic update scheme natively mitigates this by reducing the overall frequency of updates. To further guarantee stability, we adopt a *memory caching* strategy [?] that aggregates historical fast-weight checkpoints, stabilizing the parameter space to handle minute-long streams. Crucially, modeling deformation in the Synthesis Step enables memory caching to handle changing motion (Sec. 4.4). Finally, we identify that the standard dot-product inner loss used in state-of-the-art TTT [?] causes magnitude inflation over many update steps, and stabilize our framework by replacing it with an L_2 objective [4].

To demonstrate the robust memorization and generalization of Neural Space-Time Memory (NSTM), we extensively evaluate our framework on a large-scale dynamic human motion dataset, MVHumanNet++ [?]. NSTM successfully achieves stable, minute-long persistent memorization, maintaining high-fidelity synthesis of dynamic scenes while operating in real-time.

Our key contributions are summarized as follows:

- **Real-Time, Long-Context Dynamic NVS:** We propose Neural Space-Time Memory (NSTM), the first online dynamic novel view synthesis framework capable of maintaining minute-long persistent memory while operating in real-time.
- **Decoupled Memorization and Deformation:** We overcome the TTT computational bottleneck by structurally decoupling the frequencies between periodic, heavy memory updates and continuous, transformer-driven per-frame synthesis.
- **Disentangled Memory Supervision:** We introduce an alternating *memorization* and *synthesis* training regime and an auxiliary **memory loss** that explicitly forces the network to internalize persistent scene context, overcoming sparse supervision occlusion signals.
- **Memory Caching & Stability:** We enable stable, minute-long view synthesis by aggregating historical states via **memory caching**.

2 Related Works

2.1 Novel View Synthesis

The pursuit of high-fidelity novel view synthesis (NVS) has historically centered on designing robust 3D representations to establish multi-view geometric consistencies. Researchers have developed various structures, including meshes [19, 35,

43], multi-plane images [9, 13–15], neural radiance fields [30, 33, 46, 47, 51, 60, 61], point clouds [1, 45, 49, 56], 3D Gaussian splatting [21, 27, 54], etc. While effective, these representations impose implicit inductive biases that can constrain a model’s flexibility when handling complex scenes that deviate from predefined priors.

Recently, the field has shifted toward more general, data-driven architectures to reduce these geometric constraints [36, 39]. One prominent direction leverages diffusion models [20], utilizing strong 2D generative priors to hallucinate unobserved regions within a 3D-aware framework [31, 32, 38, 52, 62]. In parallel, an emerging line of research demonstrates that constructing an explicit 3D proxy is not strictly necessary at all. Frameworks such as LVSM [23] and LaCT [64] operate in a purely representation-free manner by formulating NVS entirely as a latent sequence modeling task. Instead of building spatial structures, these methods tokenize source views and utilize expressive transformer architectures to implicitly capture multi-view consistencies. By directly predicting novel views in a feed-forward manner, they successfully bypass traditional 3D structural constraints and highlight the scalability of large vision models.

2.2 Streaming Scene Reconstruction

In the context of dynamic environments, streamable novel view synthesis introduces unique complexities, requiring systems to operate causally and efficiently on continuous video streams. Unlike offline dynamic NVS methods that retrospectively optimize over the entire sequence [10, 16, 28, 29, 54, 57, 59], streaming reconstruction demands online updates using only current and historical frames under strict computational budgets. Early streaming approaches [40, 58] rely on per-frame optimization, which is prohibitively slow for real-world applications. To accelerate this process, recent work [50] incorporates optical flow to reuse past information. However, this localized tracking mechanism inevitably accumulates errors over time, leading to long-term degradation. In our work, we address this limitation by encoding a persistent memory directly into the fast weights of the network [64]. This architecture allows our model to faithfully memorize scene geometry and retain historical context, ensuring robust, temporally consistent synthesis without cumulative drift.

2.3 Memory Modules

Transformers [44] are the de-facto backbone of most recent advances in machine learning, ranging from language modeling [8, 53] to computer vision [12] and video generation [23]. In recent years, however, there has been increasing interest in designing alternative sub-quadratic architectures, due to the prohibitive quadratic cost of Transformers on long-context data. A simple but elegant solution is to replace Softmax operator with a separable kernel [26], resulting in a linear attention with a recurrent computation and linear complexity that compresses the context into a fixed-size matrix-valued memory. Building on this concept, several studies explored improving linear attention architectures with incorporating: (1)

more powerful separable kernels [2, 4, 25, 26]; (2) more expressive learning rules for better memory management such as Delta [37], Oja’s [22], and Omega [4] rules; (3) different types of forgetting (or retention) gates [6, 18, 34, 42]; and (4) novel hybrid designs of Softmax attention and subquadratic architectures [2, 7, 11].

Despite such advances, their fixed-size matrix-valued memory damages the performance of the models in retrieval-based tasks, mainly due to limited capacity of a linear matrix as the hidden state of the recurrent neural network. To this end, a new line of research starts designing methods based on “Test-time Memorization/Training” [6, 41] and more generally Nested Learning [5] that allow the use of: (1) deep memory modules [7, 41] as the state of the recurrent models, where the memory capacity is proven to exponentially grow with the depth of the memory [4]; and (2) sub-linearly growing memory modules [3, 17] based on structured matrices [17] or memory caching [3], overcoming the fixed-size effective memory of such architectures. All these architecture designs, however, are limited to 1-dimensional sequential data, and so cannot take advantages of rich space-time dependencies in video data.

3 Method

We address online novel view synthesis from dynamic video streams. At each timestep t , we observe N input views $\mathcal{O}_t = \{(\mathbf{I}_{t,i}, \mathbf{P}_{t,i})\}_{i=1}^N$, where $\mathbf{I}_{t,i} \in \mathbb{R}^{H \times W \times 3}$ is an RGB image and $\mathbf{P}_{t,i} \in SE(3)$ is the corresponding camera pose. Following LVSM [23], we tokenize the visual observations into a 1D sequence using Plücker ray embeddings. Because $\mathcal{O}_t$ provides an incomplete observation of the true dynamic scene (e.g., due to occlusion), NSTM synthesizes the target image $\mathbf{I}_t^{\text{tgt}}$ by conditioning the rendering function Φ on both the current input $\mathcal{O}_t$ and an accumulated implicit memory $\mathcal{M}_{t-1}$:

$$\mathbf{I}_t^{\text{tgt}} = \Phi(\mathbf{P}_t^{\text{tgt}}, \mathcal{O}_t; \mathcal{M}_{t-1}). \quad (1)$$

3.1 Decoupled Memorization and Synthesis Frequency

Our architecture builds upon the Test-Time Training (TTT) framework [64]. A standard block in TTT consists of a per-image self-attention module, a SwiGLU-MLP implicit memory module (f_W), and a standard feed-forward MLP. To interface with the memory, input tokens are projected into keys k_t , values v_t , and queries q_t . The memory internalizes context via a gradient-descent update on an inner loss to memorize the binding between k_t and v_t and is subsequently read via a forward query step ($o = f_W(q_t)$). More specifically, LaCT [64] optimizes a dot-product inner loss $\mathcal{L}_{\text{inner}}(f_W(k_t), v_t) = -f_W(k_t)^\top v_t$. Following a meta-learning paradigm, the overall network is trained end-to-end via an outer-loop task loss that optimizes the initial weights and projection matrices.

Naïvely applying this architecture to live video streams introduces a latency bottleneck. Standard TTT rigidly couples the memory update and query steps. To achieve high throughput, prior static frameworks process sequence chunks

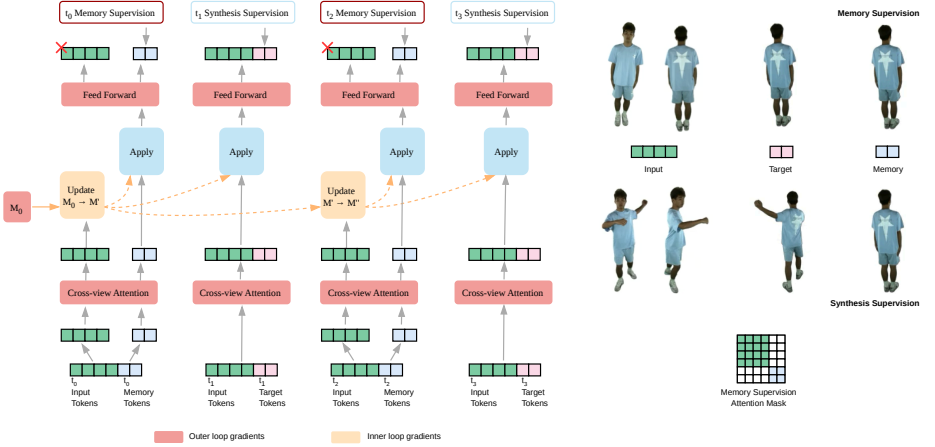


Fig. 2: NSTM Architecture and Training Regime. **Left:** To prevent the network from bypassing the memory module, we introduce an alternating training schedule. In the *Memory Supervision* step, we introduce isolated memory tokens (target camera rays) that strictly self-attend, preventing any cross-attention with the input views (see Attention Masks, bottom right). After the memory module updates from the input tokens, these isolated tokens query the fast-weights to reconstruct a target view. This forces a readout from the memory alone, ensuring the memory actually internalizes the scene context. In the *Synthesis Supervision* step, input and target tokens freely cross-attend to capture current motion, querying the previously updated memory state without altering it. **Right:** Visualizations of the input, target, and memory from these steps. Crucially, the memory readout in the synthesis step often exhibits a different pose than the current inputs. Our cross-view attention dynamically resolves this spatial deformation, successfully fusing past appearance with current motion.

in parallel. However, this relies on *chunk-level buffering*, which introduces unacceptable delays and violates the strict online constraints of live streaming. Conversely, processing the stream sequentially—by forcing a heavy gradient-based memory update at every incoming frame—incurs prohibitive computational overhead (e.g., 62ms per step; see Table 5), severely precluding real-time performance.

To overcome this computational bottleneck, NSTM decouples the frequency of memory updates from novel view synthesis, and decomposes the online inference process into two frequency-asymmetric steps (Fig. 2):

Periodic Memorization Step. Rather than updating at every frame, we strictly limit heavy gradient updates to a periodic schedule to compress novel states into the fast-weights. Crucially, the unbounded dot-product loss used by LaCT [64] can encourage increasing the magnitude of $f_W(k_t)$. We empirically observe that this causes severe magnitude inflation after repeated gradient descent over long horizons, see Tab. 2. For stability, we utilize an L_2 inner loss

$$\|v_t - f_W(k_t)\|_2^2 \quad (2)$$

alongside L_2 weight normalization, optimizing the fast-weights via the Muon optimizer [24]. Following recent implicit memory designs [4], the inner-loop learning rate η is input-dependent and learned during pretraining.

Per-frame Synthesis Step. At every frame, we bypass the expensive gradient update and execute a lightweight query against the persistent memory W' . However, because the scene is dynamic and memory updates are periodic, the memory state and the current observation are inherently misaligned (Fig. 2). Interestingly, we observe that the memory encodes a superposition of past key-value pairs, which visually manifests as a superposition of historical poses. To handle this temporal deformation, unlike the static per-image attention used in LaCT [64], we adopt a *cross-view attention* mechanism. Jointly attending across the source posed images and the target camera rays explicitly captures ongoing motion. The output then queries the memory module, dynamically fusing the current motion with historical context to synthesize the novel view at amortized real-time speeds (Fig. 5).

3.2 Disentangled Memory Supervision

A naïve joint training of the cross-view attention module and the memory module is fundamentally underconstrained. Because disocclusion signals in streaming video are naturally sparse, the model tends to over-rely on the attention and effectively bypasses historical memory. To force the persistent internalization of scene context, we introduce an alternating training regime across a sequence of T timesteps.

Memory Supervision: As illustrated in Fig. 2, on even timesteps ($t = 0, 2, \dots$), we read out from the memory and supervise this readout to reconstruct a random target view. To enforce this loss on the memory alone, it is critical to ensure we read out purely from the memory, completely isolated from the current inputs. Therefore, we introduce additional *memory readout tokens*, representing the camera rays of the target view. Crucially, we restrict the attention mask of the aforementioned cross-view attention module such that these memory tokens only self-attend amongst themselves. These isolated tokens then query the updated memory and pass through the feed-forward layers. Ultimately, they form an image, denoted as $\mathbf{I}_t^{\text{mem}}$, relying entirely on the internalized historical state. We supervise this blind retrieval via an auxiliary *memory loss*:

$$\begin{aligned} \mathcal{L}_{\text{mem}}^{(t)} &= \mathcal{L}(\mathbf{I}_t^{\text{mem}}, \mathbf{I}_t^{\text{gt}}), \\ \text{where } \mathcal{L}(\mathbf{I}, \mathbf{I}^{\text{gt}}) &= \text{MSE}(\mathbf{I}, \mathbf{I}^{\text{gt}}) + \lambda_{\text{lipps}} \text{LPIPS}(\mathbf{I}, \mathbf{I}^{\text{gt}}). \end{aligned} \quad (3)$$

λ_{lipps} is the weight to balance the LPIPS loss [63].

Synthesis Supervision: On odd timesteps ($t = 1, 3, \dots$), the network operates in standard novel view synthesis mode and we supervise the synthesized target view. Here, the standard target ray tokens freely cross-attend with the tokenized inputs $\mathcal{O}_t$ while querying the persistent memory $\mathcal{M}_{t-1}$ inherited from the immediately preceding memorization step. Consequently, despite dynamic deformations between the current observation and the previous memory state, if

a region is occluded, the network learns to dynamically fuse and recover it from this preceding memory. We supervise the output target image from this joint synthesis, denoted as $\mathbf{I}_t^{\text{synth}}$, using the identical loss formulation:

$$\mathcal{L}_{\text{synth}}^{(t)} = \mathcal{L}(\mathbf{I}_t^{\text{synth}}, \mathbf{I}_t^{\text{gt}}). \quad (4)$$

This alternating process continues until all T observations in the sequence are processed. The final training objective minimizes the joint loss across all timesteps, balancing the primary novel view synthesis task against the auxiliary memory regularization:

$$\mathcal{L}_{\text{total}} = \sum_{\substack{t=1 \\ t \text{ is odd}}}^T \mathcal{L}_{\text{synth}}^{(t)} + \sum_{\substack{t=0 \\ t \text{ is even}}}^T \mathcal{L}_{\text{mem}}^{(t)}, \quad (5)$$

where λ_{mem} controls the strength of the memory supervision.

Training-Inference Asymmetry. While we train with a 1:1 alternating ratio, our framework can execute significantly more synthesis steps than memory updates during inference. This discrepancy does not degrade performance because the synthesis steps between two memorization steps read out from the same frozen memory state and are completely independent of one another.

3.3 Memory Caching

Continuously updating the fast weights on streaming videos inevitably leads to optimization instability and memory decay (Fig. 4). To address this problem, we adapt a **memory caching** strategy recently proposed for large language models [3, 6].

Concretely, the sequential application of our periodic memory update generates a sequence of evolving memory states $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \dots$, where the ones between updates are the same. Note, at any timestep t , the active memory is $\mathcal{M}_{t-1}$ since the memory is not reset after a snapshot is taken. We average over the active memory and every K past memory states, which can be implemented recursively.

$$\bar{\mathcal{M}}_{t-1} = \frac{1}{m+1} \left(\mathcal{M}_{t-1} + \sum_{i=1}^m \mathcal{M}_{iK} \right) = \frac{1}{m+1} (\mathcal{M}_{t-1} + m\mathcal{C}_m), \quad (6)$$

$$\mathcal{C}_n = \frac{n-1}{n} \mathcal{C}_{n-1} + \frac{1}{n} \mathcal{M}_{nK} \quad (7)$$

where $m = \lfloor (t-1)/K \rfloor$. $\mathcal{C}_m$ is the average of every K past states, which we cache. In practice, we only keep one extra set of weights $\mathcal{C}_m$ to track the whole average. This aggregated state $\bar{\mathcal{M}}_{t-1}$ replaces standard memory conditioning (Eq. 1), stabilizing the parameter space and mitigating the out-of-distribution drift caused by continuous gradient updates.

Importantly, this memory caching strategy fails on standard, coupled TTT architectures. Its effectiveness fundamentally relies on our cross-view attention to explicitly handle spatial deformations, as we empirically validate in Sec. 4.4.

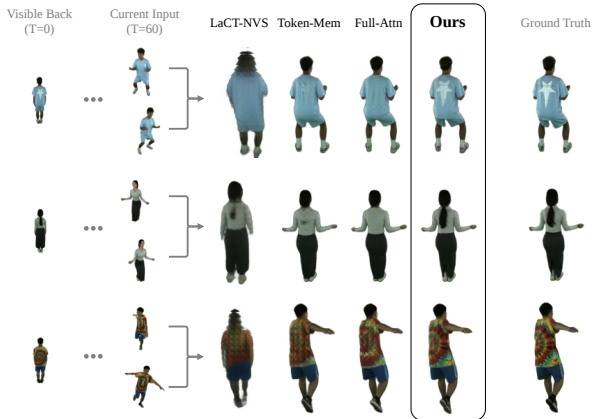


Fig. 3: MVHumanNet++ Qualitative Results. Comparison with Full-Attn (LVSM), Token-Mem (CUT3R), and LaCT-NVS after 1 minute of continuous video. NSTM accurately recalls T-shirt patterns and ponytails. Full-Attn hallucinates unseen regions, LaCT-NVS suffers OOD errors, and Token-Mem fails to recall back views.

Method	4 Timesteps					30 Timesteps					60 Timesteps				
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	mPSNR $\uparrow$	mSSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	mPSNR $\uparrow$	mSSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	mPSNR $\uparrow$	mSSIM $\uparrow$
Full-Attn	28.72	0.9659	0.0256	19.49	0.6152	28.49	0.9640	0.0271	19.08	0.5715	28.47	0.9639	0.0272	19.05	0.5694
Token-Mem	28.75	0.9638	0.0277	19.68	0.5955	28.66	0.9634	0.0284	19.42	0.5706	28.65	0.9634	0.0286	19.37	0.5674
LaCT-NVS	30.84	0.9746	0.0180	22.06	0.7348	25.67	0.9478	0.0369	16.94	0.5794	22.32	0.9275	0.0616	13.49	0.4942
Ours	30.71	0.9745	0.0186	21.78	0.7300	30.21	0.9712	0.0210	20.98	0.6646	30.09	0.9708	0.0212	20.82	0.6604

Table 1: Quantitative comparison. best, second best, and third best are colored. While LaCT-NVS significantly outperform the stateless Full-Attn in the short term ($T=4$), the standard update dot-product loss proves unstable for dynamic novel view synthesis resulting in a 38% drop in PSNR over 60 timesteps. Full-Attn fails to reconstruct any view not directly seen in the inputs as shown in Fig. 3 and therefore maintains the same performance across all timesteps. Surprisingly Token-Mem also fails to recover occlusions from memorized state and maintains a constant performance across all timesteps.

4 Experiments

We validate NSTM’s ability to maintain real-time, high-fidelity, minute-long memory under severe occlusions. We detail our experimental setup in Sec. 4.1, evaluate memory persistence and synthesis fidelity in Sec. 4.2, analyze our amortized efficiency in Sec. 4.3, and deconstruct our architecture via an ablation study in Sec. 4.4. Because PDFs cannot fully capture long-context temporal consistency, we strongly encourage reviewers to view the supplementary video and additional evaluations.

4.1 Experimental Setup

Dataset. We evaluate on MVHumanNet++ [55], featuring 60,000 multi-view motion sequences across 4,500 identities (4,400 train / 100 test). Its unconstrained motion with frequent self-occlusions makes it an ideal testbed for dy-

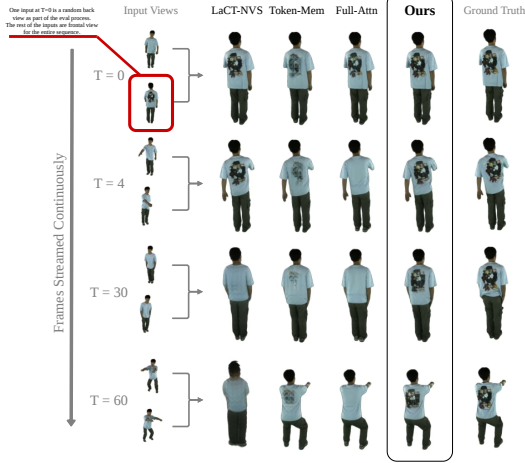


Fig. 4: Temporal memory degradation over continuous video streaming. We evaluate the synthesis quality of occluded regions across four intervals: $T = 0$, $T = 4$, $T = 30$, and $T = 60$. As the sequence progresses, baseline models quickly suffer from catastrophic drift or collapse to a stateless prior. NSTM structurally prevents this error accumulation, maintaining consistent, high-fidelity recall of the subject’s back throughout the entire sequence.

dynamic NVS. This dataset is uniquely suited for testing long-term memory tasks: the processed evaluation sequences consist of 60 frames, but these are aggressively subsampled (1 frame every 25 frames) from 50 seconds of continuous 30 FPS raw videos. We show that our approach maintains consistency across this sparse 60-frame benchmark, demonstrating NSTM’s capacity to handle minute-scale real-world temporal dynamics. Please see our supplementary materials for video results from their minute-length raw input videos.

Evaluation Protocol. To isolate true memorization from memoryless view interpolation, we design a forced-occlusion “Memory Persistence Test.” At frame 0, the model observes both the front and back views of the subject. For all subsequent frames, the back view is strictly withheld from the input stream, forcing the network to continuously retrieve the unseen back from memory. We evaluate synthesis fidelity at $T = 4$ (immediate), $T = 30$ (medium-term), and $T = 60$ (minute-scale). Note $T = 4$ is also the video length the temporal models have been trained on. To capture sustained performance rather than instantaneous peaks, standard metrics (PSNR, SSIM, LPIPS) are averaged over all frames up to that specific timestep. Furthermore, to prevent easily reconstructed static backgrounds from artificially inflating scores, we introduce mPSNR and mSSIM. Utilizing ground-truth masks, these metrics are computed strictly within the foreground human region. Additional results for a standard, unconstrained generic NVS setup are provided in the Supplementary Material.

Baselines. We compare our approach against one stateless and two stateful baselines, carefully adapted and tuned to ensure a competitive and fair evaluation. Our stateless baseline, **Full-Attn**, adapts LVSM [23] by utilizing 16 attention heads, replacing RMSNorm with LayerNorm, and omitting the standard

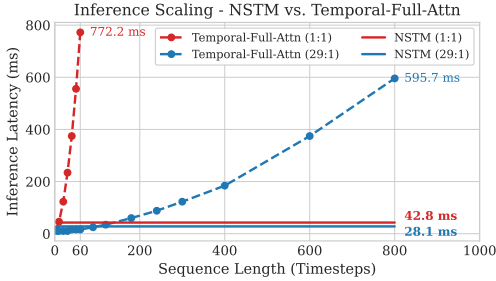
$1/\sqrt{d}$ scaling factor to optimize performance for our setting. For our stateful baselines, we first evaluate against LaCT-NVS [64], which we adapt to dynamic scenes by doing memory update and apply per-frame. We also compare against **Token-Mem**, a custom novel view synthesis model based on CUT3R [48]. Built upon DUST3R [49], Token-Mem modifies the original design by directly concatenating tokens from two source views at each timestep for the state update, entirely bypassing intermediate token supervision. The transformed tokens and target rays are then passed to a shared-weight state readout operation to predict the output RGBs. To handle continuous video, both LaCT-NVS and Token-Mem rigidly perform state updates and synthesis together as per the designs of LaCT-NVS [64] and CUT3R [48]. On high-frame-rate streams (e.g., 30 FPS), forcing an update at every single frame induces rapid catastrophic drift. Interestingly, applying our sparse 25-frame subsampling protocol to these baselines actually improves their metrics by artificially delaying this error accumulation. NSTM completely avoids this failure mode by structurally decoupling memory updates from novel view synthesis.

To ensure a fair comparison between our method and the baselines, we alternate updates and applies on consecutive frames (so an update-apply happens every two frames). The Token-Mem and CUT3R baselines perform updates every frame.

Implementation Details. We implement NSTM and all baselines in JAX. All models are trained for 500k iterations using identical optimizer hyperparameters with a total batch size of 128, utilizing 64 NVIDIA H100 GPUs. Real-time inference latency is evaluated on a single H100 GPU. During training, we dynamically construct 4-frame clips by randomly sampling temporal windows, alongside two random input views and one target view. To help the network utilize its memory module, we randomly select two completely different views as the input for the first frame. Crucially, despite training on 4 frames, NSTM successfully generalizes to 60-frame sequences during evaluation and minute-long videos in our supplementary materials, demonstrating robust long-term stability without requiring expensive long-sequence unrolling during training. Finally, for all models, the perceptual loss weighting hyperparameter is set to $\lambda_{\text{lips}} = 0.5$.

4.2 Memory Persistence and High-Fidelity Synthesis

We present our quantitative comparisons in Tab. 1 and qualitative visual evidence in Fig. 3. In the short term ($T = 4$), the advantage of explicit memory is clear: NSTM, LaCT-NVS, and Token-Mem all outperform the stateless Full-Attn, which struggles without historical context. However, the fragility of prior stateful methods becomes visible over the minute-long context ($T = 60$). Token-Mem quickly forgets historical context, its performance comparable to the stateless LVSM that never observed the back view. Similarly, LaCT-NVS suffers a severe 10.5% PSNR drop due to the instability of its continuous update mechanism. This quantitative collapse is clearly visible in Fig. 3: Full-Attn produces blurry outputs in unseen regions (e.g. the back), while LaCT suffers from out-of-distribution (OOD) distortions and noise.



Step	Latency (ms) ↓ FPS ↑	
Memorization	58.14	17.19
Synthesis	27.01	37.02

Fig. 5: Inference Performance and Scaling. (Left) Inference latency of NSTM compared to Temporal-Full-Attn based on LVSM [23]. NSTM achieves constant latency whereas full-attention results in quadratic cost. (Right) Inference performance breakdown measured on a single H100 GPU with a resolution of 256x256.

On the other hand, NSTM maintains highly robust long-term memory, degrading by a marginal 2.0% at $T = 60$. This dominance is even more pronounced in our masked foreground metrics (mPSNR/mSSIM), proving the model accurately locks in the dynamic subject’s identity rather than relying on the background. Qualitatively (Fig. 3), NSTM faithfully reconstructs specific, temporarily occluded details—such as complex t-shirt patterns and ponytails—long after they disappear from the input stream. This sustained high-fidelity recall directly validates the stabilizing effect of our L_2 objective and memory caching strategy.

4.3 Real-Time Inference Efficiency

To validate our claim of real-time, long-horizon generation, we compare the amortized inference latency of NSTM against a temporal version of Full-Attn baseline from in Sec. 4. We handle long contexts in Temporal-Full-Attn by computing dense attention across all past observations that we want to memorize. As shown in Fig. 5 this imposes a quadratic computational cost, causing latency to spike and precluding real-time rendering on continuous streams.

In contrast, NSTM structurally bypasses this bottleneck, achieving constant inference latency regardless of video length. Whether operating at a 2:1 synthesis-to-memorization ratio (our eval protocol in Tab. 1) or a 30:1 ratio (utilized for minute-long streams in our supplementary video), NSTM maintains a flat latency profile. This decoupling successfully severs the tie between sequence length and computational cost, unlocking sustained, real-time dynamic view synthesis.

4.4 Ablation Study

We validate our architectural choices in Tab. 2 and Fig. 6, evaluating the impact of each component over 60 timesteps on 390 unseen dynamic scenes from MVHumanNet [55].

Stability via L_2 Objective. We first establish a naïve causal baseline (LaCT-NVS) [64] that updates memory at every frame without decoupling memorization from synthesis. As observed, the unbounded dot-product inner loss

Method	Ablation Study (over 60 timesteps)				
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	mPSNR $\uparrow$	mSSIM $\uparrow$
LaCT-NVS Baseline	22.32	0.9275	0.0616	13.49	0.4942
LaCT-NVS w/ L_2	27.77	0.9594	0.0275	19.08	0.6189
LaCT-NVS w/ L_2 w/ Mem Caching	19.91	0.9096	0.0746	11.63	0.4509
wo/ $\mathcal{L}_{\text{mem}}$	28.23	0.9639	0.0251	19.36	0.6198
Without Mem Caching	29.24	0.9673	0.0232	19.78	0.6321
Ours w/ Mem Caching	30.09	0.9708	0.0212	20.82	0.6604

Table 2: Ablation study of our model. Metrics are averaged over 60 timesteps in 390 MVHumanNet [55] eval scenes. **best**, **second best**, and **third best** are colored.

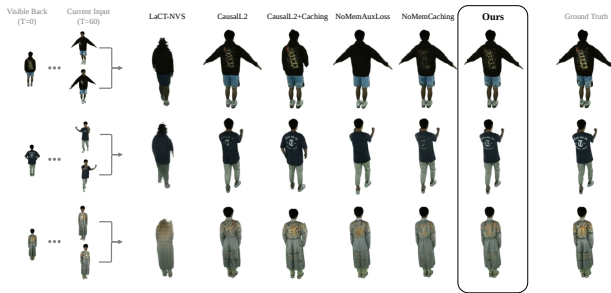


Fig. 6: Qualitative ablation study of architectural components at $T = 60$. This demonstrates the visual necessity of our decoupled memory framework and stabilization strategies. Only our full framework (Ours) stably and accurately recalls the historical back view.

causes magnitude inflation over continuous updates, resulting in poor performance (22.32 PSNR). Replacing this with a stable L_2 objective (LaCT-NVS w/ L_2) significantly improves stability and synthesis quality (+5.45 PSNR).

Memory Caching Requires Decoupled Training. While memory caching (Sec. 3.3) is designed for long-term stability, it fundamentally fails on coupled architectures. Row 3 evaluates the LaCT-NVS with L_2 baseline by performing its standard update-and-apply step at every frame, but uniquely conditioning it on the aggregated memory cache. This causes a severe performance drop (19.91 PSNR). The synthesized output becomes visually "stuck" in a historical pose. This occurs because temporal motion is inherently baked into the fast-weights; aggregating these weights creates a superposition of past poses. In our decoupled architecture, the cross-view attention explicitly learns to resolve and deform from this superimposed historical memory to match the current observation. The tightly coupled baseline lacks this explicit deformation mechanism, causing it to simply output the averaged past poses. This proves that structurally decoupling memorization from synthesis is strictly necessary to successfully aggregate historical memory.

Disentangled Supervision Enhances Memory Persistence. When we fully decouple the architecture but remove the auxiliary memory loss $\mathcal{L}_{\text{mem}}$ (No

Mem Aux Loss), performance drops to 28.23 PSNR. Without explicit pressure to reconstruct the scene independently, the network takes a shortcut, over-relying on the cross-view attention’s current visual context and partially collapsing to a stateless prior. Introducing $\mathcal{L}_{\text{mem}}$ forces robust geometric internalization (+1.01 PSNR).

Long-term Stability via Memory Caching. Finally, applying our memory caching strategy to the fully decoupled and properly supervised model (Ours full) yields our highest performance (30.09 PSNR), successfully mitigating long-term weight drift and preserving minute-long context.

5 Discussion

5.1 Limitations and Future Work

Our method performs online updates which are spaced out by fixed time intervals. We note that the updates may miss events that happen between the updates. To avoid this issue, one could perform large chunk updates, which essentially perform the update step on multiple frames at a time, each with their own time embedding. This change would also enable the model to perform updates with irregular frame spacing (e.g. unsynced cameras on a multi-view rig). In general, this paper leaves open the question of which frames should be memorized. Future work could explore the tradeoff between more frequent updates and larger-chunk "batch" updates.

Models like ours will eventually hit capacity limits and forget past observations. To improve persistence and stability, the model could learn scales for the weights in the memory cache in a similar fashion to other methods [3, 65].

5.2 Conclusion

In this work, we presented Neural Space-Time Memory (NSTM), a novel framework that bridges the gap between long-context memory and computational efficiency for dynamic novel view synthesis. By disentangling periodic memory updates from per-frame transformer-based deformation modeling, and introducing a targeted memory auxiliary loss, NSTM successfully internalizes persistent scene context while maintaining real-time speed. Furthermore, our memory caching strategy and stable L_2 update rule enhance long-term stability, unlocking minute-long memory horizons and achieving state-of-the-art results compared to existing baselines.

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